H1-REGULARITY IS INHERITED BY A BLOCK OF THE SEQUENCE MODULO THE IDEAL GENERATED BY THE EARLIER BLOCK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- H_1-regularity of a sequence expressed through the trivial relations among its elements. definition and reformulation in the same section, source lines 7100-7298, before the counted statement. Supplies the goal of the proof in the form of a decomposition of relations, both for the long sequence and for the quotient sequence.
- An element of the ideal generated by f_1 to f_n is a combination of those elements. used at proof lines 7629-7634 where the relation in A/I is lifted to a relation in A. Produces the lift of the relation, which is what makes the hypothesis on the long sequence applicable.
- The quotient map A to A/I sends the trivial relations of the long sequence to trivial relations of the quotient sequence. used at proof lines 7635-7637. Transports the decomposition found in A to the decomposition required in A/I, closing the proof.

Source reference. Original lemma 068L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8010–8010. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let A be a ring. Let $f_1, \ldots, f_n, g_1, \ldots, g_m \in A$ be an H_1-regular sequence. Then the images $\overline{g}_1, \ldots, \overline{g}_m$ in $A/(f_1, \ldots, f_n)$ form an H_1-regular sequence.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Set $I = (f_1, \ldots, f_n)$. We have to show that any relation $\sum_{j=1,\ldots,m} \overline{a}_j \overline{g}_j$ in A/I is a linear combination of trivial relations. Because $I = (f_1, \ldots, f_n)$ we can lift this relation to a relation

$$\sum\nolimits_{j=1,\ldots,m} a_j g_j + \sum\nolimits_{i=1,\ldots,n} b_i f_i = 0$$

in A. By assumption this relation in A is a linear combination of trivial relations. Taking the image in A/I we obtain what we want. $\square$